Symmetries in Physics — Exercise Sheet 4

Irreducibility, averaging, and Schur's lemma

Variant: Questions only

Exercise 1. Let $C_3 = \langle r \rangle$ act on $\mathbb{C}^2$ by $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$.

(i) Check $A^3 = \mathbb{1}$, and show that A is not unitary for the standard inner product.

(ii) Average the standard inner product over the group, and check that the Gram matrix M of the result satisfies $A^\dagger M A = M$.

(iii) Let $\omega = e^{2\pi \mathrm{i}/3}$. Check that $v = (1, -\omega)$ and $v' = (1, -\omega^2)$ are eigenvectors of A, so that $\mathbb{C}v$ and $\mathbb{C}v'$ are invariant lines, and explain why they are the only ones. Show that v and v' are orthogonal for the invariant product but not for the standard one. Which result of Lecture 4 made this inevitable?

(iv) Apply Gram–Schmidt to (e_1, e_2) in the invariant product, and write A in the resulting orthonormal basis. Which matrix of Lecture 3 do you get?

Exercise 2. Maschke's theorem has two hypotheses. This exercise shows what each is for.

(i) Take the representation $D(n) = \left(\begin{smallmatrix} 1 & n \\ 0 & 1 \end{smallmatrix}\right)$ of the infinite group $(\mathbb{Z}, +)$ from Lecture 4's example "Reducible but indecomposable". Which step of the averaging proof cannot be carried out? Prove that no inner product on $\mathbb{C}^2$ makes ρ unitary. *Hint*: use the corollary on orthogonal complements.

(ii) Now let the numbers be *bits*: vectors in $V = \{0, 1\}^2$, added and scaled modulo 2. Let $C_2 = \{e, a\}$ act on V by swapping coordinates. Show that the line spanned by $(1, 1)$ is invariant but has no invariant complement, and say where the averaging proof breaks here.

Exercise 3. Lecture 4's worked example ends with a remark: the standard representation of S_3 and the defining representation of D_3 are equivalent. Prove it. Here is everything you will need.

- *The standard representation* (Lecture 4). A permutation σ acts on $\mathbb{C}^3$ by $\rho(\sigma)e_k = e_{\sigma(k)}$. The subspace $V_0 = \{\sum_i v_i = 0\}$ is spanned by $f_1 = e_1 - e_2$ and $f_2 = e_2 - e_3$, and in the basis (f_1, f_2) the matrix of the cycle is

$$B\big((1\,2\,3)\big) = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

  which is the matrix A of Exercise 1.

- *The defining representation* (Lecture 3). The generators r (the rotation through $2\pi/3$) and s (the reflection in the y-axis) are represented by

$$D(r) = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \qquad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

  and the vertices of the triangle are $p_1 = (0, 1)$, $p_2 = (-\frac{\sqrt{3}}{2}, -\frac{1}{2})$ and $p_3 = (\frac{\sqrt{3}}{2}, -\frac{1}{2})$.

- *The isomorphism* $D_3 \cong S_3$ (Lecture 1): $r \mapsto (1\,2\,3)$, $s \mapsto (2\,3)$. It assigns to each element the permutation by which that element permutes the vertex labels.

(i) Compute $B\big((2\,3)\big)$.

(ii) Find an invertible T with $T\,B\big((1\,2\,3)\big) = D(r)\,T$ and $T\,B\big((2\,3)\big) = D(s)\,T$. By Sheet 3, Exercise 6, checking the two generators is enough. *Hint*: send e_k to the vertex p_k of the triangle.

(iii) Use Schur's lemma to show that every T satisfying both equations of (ii) is a multiple of the one you found.

(iv) Explain without computing why the standard inner product of $\mathbb{C}^3$, restricted to V_0, is invariant under all of S_3. Compute its Gram matrix G in the basis (f_1, f_2), and compare it with the matrix M of Exercise 1.

Exercise 4. An atom sits at the centre of the square of Sheet 3, Exercise 2. Its orbitals $2p_x = x_1\, f(|\vec{r}\,|)$ and $2p_y = x_2\, f(|\vec{r}\,|)$, with one radial function f, transform like x_1 and x_2, because the symmetries do not change $|\vec{r}\,|$. So by that exercise they carry the defining representation of D_4, with $D(r) = R(\pi/2)$ and $D(s) = \mathrm{diag}(1, -1)$.

(i) Suppose that a line $\mathbb{C}v$ were invariant. Show that $\rho(g)v = \chi(g)v$ defines a one-dimensional representation χ of D_4, and use Sheet 3, Exercise 1 and the matrix $D(r^2)$ to reach a contradiction. Conclude that the representation is irreducible.

(ii) Let H be a Hamiltonian that maps the span of $2p_x$ and $2p_y$ to itself and commutes with every $\rho(g)$ there. Show that $2p_x$ and $2p_y$ have the same energy.

(iii) Now stretch the square into a rectangle that is not a square, with its sides along the axes. Its symmetries are e, r^2, s and r^2s: the Klein four-group of Sheet 1, Exercise 3. Explain without computing why the representation on the span of $2p_x$ and $2p_y$ must now split into one-dimensional representations. Find the one carried by $2p_x$ and the one carried by $2p_y$. Suppose the Hamiltonian of the stretched system still maps the span to itself and commutes with the four symmetries of the rectangle. Does symmetry still force the two energies to be equal?

Exercise 5. Let ρ be a complex irreducible representation of a finite group G on V.

(i) Use Schur's lemma to show that every central element z acts as a scalar, $\rho(z) = \lambda(z)\mathbb{1}$.

(ii) Suppose moreover that ρ is faithful. Show that $z \mapsto \lambda(z)$ is an injective homomorphism from the centre $Z(G)$ to $\mathbb{C} \setminus \{0\}$, and deduce that $Z(G)$ is cyclic.

(iii) Conclude that the Klein four-group, the symmetry group of the rectangle in Exercise 4, has no faithful irreducible representation.

Exercise 6. The quaternion group is $Q_8 = \{\pm 1, \pm \mathrm{i}, \pm \mathrm{j}, \pm \mathrm{k}\}$ with $\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = \mathrm{ijk} = -1$.

(i) Find all one-dimensional representations. *Hint*: argue as in Sheet 3, Exercise 1.

(ii) Realise a two-dimensional representation with the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

sending $\mathrm{i}, \mathrm{j}, \mathrm{k}$ to $-\mathrm{i}\sigma_1$, $-\mathrm{i}\sigma_2$ and $-\mathrm{i}\sigma_3$.

(iii) Show that a reducible representation of a finite group always has a non-scalar operator that commutes with all of its matrices. *Hint*: by Maschke's theorem, project onto an invariant subspace along an invariant complement.

(iv) Show that only the scalar matrices commute with $-\mathrm{i}\sigma_1$, $-\mathrm{i}\sigma_2$ and $-\mathrm{i}\sigma_3$, and conclude that the representation of (ii) is irreducible.

(v) The representation of (ii) is faithful. Check Exercise 5(ii) on it.